



$S_0 = 0$
 $S_1 = 1$
 $S_2 = 10$
 $S_3 = 101$

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$S_4 = 1010 \rightarrow 10100$
 10101

ECE270: ELD: Quiz 2A (20 Minutes)

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1. Write a Verilog code to detect the sequence 1010 (Overlapping). (8 Marks)

Field programmable gate array

2. Explain the difference between FPGA and CPLD. (2 marks)

complex programmable logic device

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1. module sequence-detector (input in_0, in_1, reset, clk_pb,
    output detect, [2:0] fsm);
    reg [2:0] present-state = 0;
    reg [2:0] next-state;
    always @ (posedge clk_pb or posedge reset)
    begin
        if (reset)
            present-state <= 0;
        else
            present-state <= next-state;
    end

    always @ (*)
    begin
        fsm <= present-state;
    end

    assign detect =
    reg out1;
    always @ (*)
    begin
        if (present-state == 3'b100)
            out1 <= 1;
        else
            out1 <= 0;
    end

```

assign data = out1;

always @ (*) begin

case (prmt - state)

3'b000: if (in-1)

next-state <= 3'b001;

else

next-state <= 3'b000;

3'b001: if (in-1)

next-state <= 3'b001;

else

next-state <= 3'b010;

3'b010:

if (in-1)

next-state <= 3'b011;

else

next-state <= 3'b000;

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3'b011:

if (in-1)

next-state <= 3'b001;

else

next-state <= 3'b100;

3'b100:

if (in-1)

next-state <= 3'b000;

else

next-state <= 3'b011;

default next-state <= 3'b000;

endcase

end

endmodule

assign data = out1;

2. FPGA is reprogrammable device that uses LUTs to perform functions

1. CPLD is reprogrammable device with multiple PLDs connected by interconnect matrix